

University of Wolverhampton

Department of Computing and Mathematical Sciences

Master Thesis

**AI-Driven Models for Short-Term 5G
Traffic Prediction: A Performance Study**

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Abstract

The advent of 5G networks has introduced unparalleled advancements in connectivity, enabling high-speed data transfer and latency-sensitive applications. However, the dynamic and nonlinear nature of 5G network traffic presents challenges in resource management and service optimization. This study investigates the application of Gated Recurrent Units (GRU), Long Short-Term Memory (LSTM), and Support Vector Machines (SVM) for short-term traffic prediction in 5G networks. By analyzing real-world traffic data, the research evaluates model performance using metrics such as Root Mean Squared Error (RMSE) and R-squared (R^2). Findings reveal that GRU achieved the highest predictive accuracy followed closely by LSTM and SVM, demonstrating comparable effectiveness. These results underscore the potential of AI-driven approaches for enhancing real-time network management. The study concludes with recommendations for hybrid models and further incorporation of external factors to improve predictive accuracy and scalability in next-generation networks.

1 Introduction

The evolution of telecommunication networks has been a cornerstone of global communication, with advancements from the 1st Generation (1G) to the 5th Generation (5G) shaping the way people connect and share information (Sharma, 2013). Each generation has addressed specific user needs, enhanced capabilities and overcoming limitations. The introduction of 1G allowed for mobile phone usage but was constrained to basic voice services with suboptimal quality. With 2G (2nd Generation), significant improvements were made, including clearer voice transmission, increased security, and the ability to transmit mobile data for text messaging and multimedia sharing. However, its data rates, approximately 0.2 Mbps, were inadequate by modern standards. The advent of 3G (3rd Generation) marked a pivotal moment, enabling much faster data speeds of up to 2 Mbps, thus facilitating real-time video calls, internet browsing, and video streaming (Rastvortseva and Kamenewa, 2024).

The leap to 4G (4th Generation) brought unprecedented speeds, supporting high-definition video streaming and online gaming. By 2023, LTE (Long-Term Evolution) networks like Emtel Mauritius achieved download speeds nearing 55 Mbps (Toan and Truong, 2021), illustrating the upper thresholds of 4G technology. However, with applications becoming increasingly sophisticated and data-intensive, even these speeds became insufficient. This gave rise to 5G networks, designed to support advanced applications such as machine-to-machine communication and remote robotic control with minimal latency. In 2023, South Korea set a global record with 5G download speeds of 432 Mbps, showcasing the immense potential of this technology (MacMillan *et al.*, 2023). As of 2027, 5G is expected to serve 4.4 billion subscribers worldwide, according to Ericsson. This exponential growth not only underscores the importance of 5G but also highlights the challenges it brings, such as maintaining service quality amidst rising traffic loads.

The dynamic and nonlinear nature of network traffic in 5G systems poses significant challenges for predictive modelling. While long-term traffic forecasting has been extensively studied, primarily for network design and capacity planning, short-term predictions are gaining prominence. These predictions, focused on intervals ranging from minutes to hours, are critical during events like holidays or emergencies where traffic spikes are unpredictable. Accurate short-term forecasting enables network operators to allocate resources dynamically, optimize traffic flow, and enhance the Quality of Service (QoS), thereby mitigating risks of service disruptions.

Artificial intelligence (AI) techniques are increasingly being explored to address these challenges. Among the most prominent are Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), and Support Vector Machines (SVMs). LSTMs are renowned for their ability to model long-term dependencies in sequential data, making them effective for capturing trends in traffic patterns. GRUs, a simplified variant of LSTMs, offer comparable performance with reduced computational overhead, making them particularly suited for real-time applications. Meanwhile, SVMs excel in handling small datasets and nonlinear relationships, providing an alternative approach to deep learning methods.

Despite these advancements, a noticeable gap exists in comparative studies focusing on GRU, LSTM, and SVM models specifically for short-term 5G traffic prediction. This research aims to bridge these gaps by conducting a comprehensive comparative analysis of GRU, LSTM, and SVM models for short-term traffic forecasting in 5G networks. It will evaluate these models using metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared values across diverse traffic scenarios. Furthermore, the study will investigate the role of preprocessing techniques, such as normalization and noise reduction, in improving model performance.

The study also seeks to analyze the computational efficiency of GRUs relative to LSTMs and SVMs in real-time applications and determine optimal configurations, such as look-back window sizes and hyperparameter settings, for each model.

By addressing these objectives, this research aims to provide actionable insights for telecommunication operators, contributing to the broader field of predictive modeling in next-generation networks. The findings are expected to inform the development of adaptive systems capable of managing the complexities of 5G environments effectively and efficiently.

1.1 Research Questions

The behavior of network traffic is fundamentally nonlinear and highly variable, presenting substantial challenges for predictive modeling in modern telecommunication systems. With the rapid proliferation of 5G networks and an exponential increase in subscriber numbers, gaining insights into traffic patterns has become essential. Accurate short-term traffic prediction offers critical insights into network behavior, enabling service providers to identify potential issues proactively and optimize resource distribution. Such preemptive strategies can significantly mitigate the risk of sudden service disruptions, ensuring a smooth user experience.

Numerous studies have investigated models for telecommunication traffic prediction, with Long Short-Term Memory (LSTM) networks and Auto-Regressive Integrated Moving Average (ARIMA) models gaining prominence in this domain. However, these models primarily cater to long-term forecasting, excelling at detecting overarching trends and periodic fluctuations. While LSTM networks adeptly model long-term dependencies and ARIMA applies statistical principles to predict stable traffic environments, both exhibit limitations when addressing the rapid changes and bursts typical of short-term predictions in 5G networks.

Gated Recurrent Units (GRUs) have emerged as a compelling alternative due to their streamlined architecture and reduced computational requirements compared to LSTMs. GRUs have shown comparable or even superior performance in certain scenarios, particularly those requiring real-time responses. Additionally, Support Vector Regression (SVR), renowned for its capability to manage small datasets and model nonlinear behaviors, provides a complementary approach to neural networks. Despite these advancements, there is a noticeable gap in research comparing the efficacy of GRUs, LSTMs, and SVR for short-term traffic forecasting in the highly dynamic 5G environment.

This study aims to fill existing research gaps by conducting a comprehensive comparative analysis of GRU, LSTM, and SVR models for short-term 5G traffic prediction. It will evaluate these models across varied traffic scenarios using metrics like RMSE, MAE, and R-squared values. Furthermore, the investigation will explore the role of preprocessing techniques, such as normalization and noise reduction, in improving forecasting precision. This study will address the following questions:

- How do GRU, LSTM, and SVR models perform in short-term traffic forecasting for 5G networks, and what are their respective advantages and limitations?
- How does the computational efficiency of GRUs compare with that of LSTMs and SVR in real-time traffic forecasting?

The aims and objectives of the study are:

- To evaluate the performance of GRU, LSTM, and SVR models in short-term traffic forecasting for 5G networks, identifying their strengths and limitations.
- To analyze the computational efficiency of GRUs in comparison to LSTMs and SVR for real-time traffic forecasting.

1.2 Outline of Thesis

The thesis explores the application of AI models, specifically GRU, LSTM, and SVR, for short-term traffic prediction in 5G networks, addressing the challenges posed by dynamic and nonlinear traffic patterns. Chapter 1 is an introduction to telecommunication advancements and the importance of predictive modeling, followed by a review of existing methodologies and their limitations in Chapter 2. The methodology in Chapter 3 details the dataset, preprocessing techniques, and model configurations used in the study, while the results and analysis in Chapter 4 compare model performance using metrics like RMSE and R^2 . The conclusion in Chapter 5 synthesizes key insights and emphasizes the significance of the study in advancing predictive modelling for next-generation networks.

2 Literature Review

(Jaffry and Hasan, 2020) investigate the potential of recurrent neural networks (RNNs) for predicting traffic patterns in cellular networks. The research highlights the increasing complexity of cellular traffic and emphasizes the importance of advanced models like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks. These models, designed to handle sequential data and mitigate the vanishing gradient problem, are compared against Feed-Forward Neural Networks (FFNN) and traditional statistical methods like ARIMA.

FFNN, despite its simplicity, demonstrated moderate accuracy with large datasets but struggled to converge effectively when training data was limited. This limitation underscores its inability to capture temporal dependencies inherent in cellular traffic. In contrast, LSTM and GRU networks excelled in capturing non-linear patterns and long-term dependencies, with GRU showing a slight edge in computational efficiency. The study also benchmarked ARIMA as a traditional statistical baseline. While ARIMA performed adequately on large datasets, it failed to model non-linear and abrupt traffic changes, making it less suitable for dynamic and sequential environments like cellular networks.

Performance was evaluated using metrics such as RMSE and MAE. Experimental results revealed that both LSTM and GRU outperformed FFNN and ARIMA, particularly when handling smaller datasets or more complex traffic patterns. The findings demonstrate that while FFNN and ARIMA provide useful baselines, neural networks like GRU and LSTM are better suited for short-term traffic prediction in cellular networks. These insights pave the way for future work on hybrid approaches and real-time applications in 5G network environments.

The application of LSTM networks for predicting mobile network traffic directly from raw LTE control channel data was examined by (Trinh, Giupponi and Dini, 2018). This research aims to address the challenge of modelling complex and dynamic traffic patterns in mobile networks, a critical aspect of optimizing resource allocation and improving network performance.

The study uses raw data from LTE networks, specifically collected from eNodeBs, and emphasizes multi-step prediction for traffic patterns. The authors propose a deep LSTM-based architecture designed to capture temporal dependencies inherent in mobile traffic. The architecture leverages multiple stacked LSTM layers to sequentially extract temporal features, enabling the model to predict mobile traffic for up to five time steps into the future. The model's performance is measured using the Normalized Root Mean Square Error (NRMSE) metric, which allows a robust evaluation of predictive accuracy.

Experimental results demonstrate that the LSTM network significantly outperforms traditional methods like ARIMA in handling the non-linear and oscillatory nature of mobile traffic. While ARIMA is effective in stable traffic scenarios, it struggles with the dynamic variations typical of mobile network data, often converging to average traffic values rather than capturing fluctuations. The LSTM model, on the other hand, excels at maintaining prediction accuracy across multiple time steps, with only a gradual decline in performance as the prediction horizon increases. The study also highlights challenges, such as the increased error margin for long-term predictions and the computational complexity associated with deep learning models.

The findings affirm the effectiveness of LSTM networks for short-term mobile traffic prediction, making them a valuable tool for real-time traffic forecasting in cellular networks. Future research is suggested to integrate multivariate factors, such as user mobility and network context, and to explore hybrid models for improved performance. This paper contributes to advancing AI-driven solutions for adaptive traffic management in mobile networks.

In another study, (Yadav *et al.*, 2021) investigates the utilization of mobile traffic data to derive insights into urban dynamics and mobility. By analyzing extensive datasets collected from mobile networks, the authors aim to enhance the understanding of urban traffic behavior and provide a framework for smart city applications.

This study applies advanced machine learning techniques to massive mobile data sets, identifying patterns and relationships within the data. The approach includes preprocessing steps such as normalization and feature extraction, followed by the application of models like LSTM networks. LSTMs are employed to capture temporal dependencies and non-linear trends in the data, which are crucial for understanding mobile traffic dynamics over time.

The findings indicate that LSTM models outperform traditional statistical methods and baseline machine learning approaches in predicting mobile traffic patterns. The model's ability to capture long-term dependencies enables it to effectively model the temporal and spatial complexities inherent in urban traffic data. The results demonstrate that the LSTM model achieves lower prediction errors compared to methods such as ARIMA, highlighting its suitability for handling the dynamic nature of urban environments.

The study contributes to the domain by bridging mobile traffic analysis with urban planning, proposing a predictive framework that supports resource allocation and mobility management in smart cities. It also emphasizes the importance of integrating spatial and temporal features into predictive models to improve their accuracy. The study concludes by recommending the exploration of hybrid models and the inclusion of real-time data streams to enhance prediction capabilities, paving the way for adaptive and intelligent urban systems.

(Fang, Zhao and Yang, 2022) explore the combination of wavelet packet denoising (WP) and deep Gaussian processes (DGP) for enhancing the accuracy of 5G network traffic prediction. With the increasing complexity and noise in network traffic data, traditional methods have struggled to maintain prediction stability and precision. This study aimed to address these challenges by leveraging WP for denoising and DGP for robust nonlinear learning.

The proposed WP-DGP model employed a Gaussian radial basis function (RBF) kernel and comprises three hidden layers optimized using an Expectation-Maximization (EM) algorithm over 14 iterations. The dataset, split in a 7:3 ratio for training and testing, was benchmarked against ARIMA, BP neural networks, and LSTM models using error metrics such as RMSE, MAPE, and MAE. Results demonstrate the WP-DGP model's superior performance, with a RMSE of 3.81 and MAPE of 1.24%, outperforming ARIMA, BP, and LSTM, which showed higher RMSEs of 6.86, 8.04, and 8.82, respectively. The findings demonstrate that the WP-DGP model significantly outperforms the other methods, achieving the lowest RMSE and MAPE scores, which underscores its robustness and accuracy in handling noisy and volatile data. The study concludes that combining WP with DGP provides an effective framework for network traffic prediction, addressing the limitations of both traditional and deep learning methods. Future work includes integrating additional contextual factors and optimizing the computational efficiency of DGP for real-time applications in 5G environments. This research contributes to the growing field of intelligent network management by proposing a highly accurate and resilient predictive approach.

The application of LSTM and GRU models in predicting traffic flow patterns in urban networks was researched by (Fu, Zhang and Li, 2016). The research compares these advanced neural networks against traditional methods such as ARIMA, emphasizing their suitability for sequential data with non-linear characteristics.

LSTM and GRU models were trained using historical traffic data with multiple hidden layers to capture temporal dependencies effectively. The dataset, divided into training and testing sets, represented real-world traffic scenarios, including fluctuating and oscillating patterns.

Experimental findings demonstrated that both LSTM and GRU significantly outperformed ARIMA in capturing short-term traffic dynamics. The GRU model exhibited a marginally lower computational overhead compared to LSTM while maintaining comparable predictive accuracy, making it preferable for scenarios requiring computational efficiency. The ARIMA model, though effective in stable conditions, struggled with rapid fluctuations, resulting in higher prediction errors.

For performance, the LSTM model achieved an RMSE of 0.8 and an MAE of 0.5 on the test set, while GRU yielded similar metrics with slightly faster convergence during training. In contrast, ARIMA recorded an RMSE of 1.2 and an MAE of 0.9, reflecting its limitations in non-linear and volatile environments.

The paper highlights the potential of LSTM and GRU networks for real-time traffic prediction, especially in dynamic urban settings. It concludes by recommending further research into hybrid models that integrate spatial and temporal features for enhanced accuracy. The findings underscore the transformative role of deep learning in urban traffic management and smart city applications. This study contributes to advancing predictive modelling by validating neural networks' capabilities in handling complex temporal datasets.

The implementation of LSTM networks for real-time traffic prediction in 5G networks was also applied in (Kumar *et al.*, 2024), focusing on enhancing network management and optimization. By leveraging LSTM's sequential learning capabilities, the study addresses the dynamic and nonlinear nature of 5G traffic. The authors utilize a dataset divided temporally into training, validation, and test sets to prevent data leakage.

Experimental results highlight the model's performance across various prediction horizons. For a 15-minute horizon, the RMSE is 185.3, MAE is 120.5, and R-squared is 0.78, reflecting strong predictive accuracy. However, as the prediction horizon extends to 30 minutes and 1 hour, the RMSE increases to 220.6 and 240.9, respectively, while R-squared decreases to 0.73 and 0.69. These findings underline the model's robustness for short-term predictions but reveal limitations for longer horizons due to increasing complexity and variability in traffic patterns.

The study demonstrates that LSTM networks effectively capture temporal dependencies and nonlinear relationships, facilitating real-time predictions. The high R-squared values indicate that the model explains a significant portion of the variability in traffic volume, providing critical insights for network operators. Despite these strengths, the authors acknowledge that factors such as weather conditions, events, and road availability, which influence traffic behavior, are not included in the model. They recommend integrating contextual parameters and exploring advanced techniques like ensemble learning and cross-model learning to improve accuracy. The proposed LSTM-based model shows significant promise for real-time traffic prediction in 5G networks, offering a foundation for intelligent network management.

(Tuna and Soysal, 2021) investigate the application of LSTM and GRU models for predicting traffic volume in live 5G network environments. The study emphasizes the importance of accurate traffic prediction in addressing the complexities arising from integrating diverse technologies in 5G and beyond networks.

Using multivariate data from eNodeB network nodes, the authors trained LSTM and GRU models to forecast traffic load variations. They explored how these models perform in capturing trends during peak hours and responding to sudden traffic spikes or drops. The evaluation metrics include MAE and RMSE,

with results showing LSTM achieving lower errors in stable conditions while GRU excelled in dynamic scenarios involving rapid changes in traffic.

Key findings highlight GRU's superior efficiency in scenarios with abrupt traffic variations due to its simpler architecture and fewer parameters compared to LSTM. However, both models showed limitations in accurately forecasting extreme fluctuations, occasionally predicting higher or lower resources than required. The study also noted the significance of using multivariate datasets, which enhanced the predictive performance by incorporating additional contextual variables.

The authors conclude that while both LSTM and GRU models offer promising solutions for traffic prediction in 5G networks, further work is required to enhance their ability to generalize across different network conditions. They recommend exploring ensemble models and integrating external factors like weather or mobility patterns to improve accuracy.

In (Nayak, 2021) more insight was provided into real-time 5G traffic prediction using time series analysis and machine learning models, particularly focusing on advanced deep learning architectures. Central to its methodology is the application of LSTM models, which are optimized for sequence prediction tasks. The study highlights the effectiveness of LSTM models in leveraging sequential learning for predicting 5G network traffic, overcoming challenges related to nonlinearity, burstiness, and fluctuations in traffic data.

The dataset used in the research encompasses traffic metrics gathered from various urban and suburban 5G network deployments. Key preprocessing steps involved converting raw data into usable formats, ensuring timestamp alignment, and handling anomalies. LSTM models were trained using a multistep prediction methodology, where both temporal and spatial correlations were analyzed.

For short-term prediction intervals, the LSTM model demonstrated superior accuracy, with an RMSE of 185.3 for a 15-minute horizon and a slightly reduced accuracy for longer prediction horizons. An R-squared value of 0.78 was achieved, indicating that approximately 78% of the variance in traffic data was explained by the model during this interval. For longer horizons, the performance decreased slightly, with R-squared values dropping to 0.73 for 30 minutes and 0.69 for a 1-hour prediction horizon. This reflects LSTM's strength in capturing temporal dependencies for real-time predictions. However, its performance slightly declined for longer intervals due to compounding errors inherent in multistep forecasting.

(Chi and Shi, 2018) investigate short-term traffic flow forecasting using a hybrid ARIMA-SVM algorithm, focusing on addressing the limitations of traditional forecasting methods. The authors argue that the nonlinear, dynamic, and time-varying nature of traffic flow renders conventional models like ARIMA insufficient, necessitating the integration of machine learning techniques such as SVM for enhanced accuracy.

The proposed ARIMA-SVM model operates in two stages: ARIMA is employed to predict the linear components of traffic flow, and its residual errors are modeled using SVM to capture nonlinear dependencies. The dataset used for the study comprises 285 traffic flow data points from a specific road in Ji'nan, China, spanning a single day. The data is split into 200 training samples and 85 testing samples. SVM hyperparameters are optimized using 10-fold cross-validation, with the radial basis function selected as the kernel due to its suitability for nonlinear data.

The hybrid ARIMA-SVM model achieves superior results compared to standalone ARIMA and SVM models, with an MAE of 12.64, MSE of 274.15, and MAPE of 11.68%. These values underscore the hybrid model's capacity to effectively combine the strengths of linear and nonlinear modeling, resulting in improved prediction accuracy. Additionally, the ARIMA-SVM model outperformed a neural network with five hidden neurons, indicating its robustness and efficiency for small-sample scenarios.

The study concludes by highlighting the innovative nature of using ARIMA-SVM for traffic flow prediction and emphasizes its potential application in intelligent transportation systems. However, the authors acknowledge limitations, such as the lack of external factors like weather or events influencing traffic flow.

Another study by (Liu, 2021) analyses the effectiveness of Support Vector Regression (SVR) in short-term traffic flow prediction. The research addresses urban congestion issues by leveraging SVR's ability to manage nonlinear and dynamic traffic data. The study utilizes traffic flow data collected from Xuancheng, Anhui Province, over a week, focusing on peak hours for detailed analysis.

The authors compare SVR with other forecasting models, including ARIMA, LSTM, and Backpropagation Neural Network. SVR achieves superior performance with an RMSE of 14.97, MAE of 9.38, and MAPE of 0.64%. ARIMA shows comparable results in terms of RMSE but exhibits higher errors in MAE and MAPE, indicating a less balanced performance. LSTM and BP models demonstrate greater flexibility in capturing nonlinear patterns but fall short due to overfitting and increased computational demands, particularly BP, which requires more training iterations.

The SVR model's kernel parameters were tuned to optimize its predictive capability. Using a radial basis function kernel, the SVR model balances accuracy and complexity, avoiding overfitting while ensuring generalization. The training phase focused on daily patterns, allowing the model to account for the inherent variability of weekday and weekend traffic flows.

The study concludes that SVR is highly effective for short-term traffic forecasting, outperforming more complex models in scenarios with limited data. However, it acknowledges the potential to enhance accuracy by incorporating external factors such as weather and events. This research underscores the applicability of SVR in intelligent transportation systems, providing urban planners with actionable insights to mitigate congestion and optimize traffic resource allocation.

(Wang *et al.*, 2021) addresses the challenges of short-term traffic prediction by proposing a hybrid SVR-LSTM model tailored for time-sensitive and dynamic traffic environments. The hybrid model overcomes the limitations of traditional methods like ARIMA by integrating SVR and LSTM, enabling more accurate forecasting for both linear and non-linear trends.

The research utilizes real-world traffic data, pre-processed to handle missing values and noise. SVR is applied to capture linear dependencies in the dataset, while LSTM models are designed to learn temporal patterns. The hybrid model combines these outputs to predict traffic flow, effectively addressing the diverse characteristics of traffic dynamics. Key evaluation metrics include RMSE and MAPE, providing a comprehensive assessment of prediction accuracy.

Experimental results reveal that the SVR-LSTM hybrid model significantly outperforms individual SVR and LSTM models. RMSE and MAPE scores for the hybrid approach demonstrate lower errors compared to its standalone counterparts. For instance, the hybrid model achieves an RMSE reduction of approximately 15% over LSTM and 20% over SVR, highlighting its effectiveness in short-term traffic prediction.

(Zeng *et al.*, 2008) explores the efficacy of an Accurate Online Support Vector Regression (AOSVR) model for real-time traffic forecasting. The research acknowledges the limitations of traditional methods, such as neural networks and static Support Vector Machines (SVMs), in adapting to dynamic traffic environments. By leveraging an online learning strategy, the proposed AOSVR model dynamically updates its forecasting function as new traffic data arrives, making it suitable for real-time Intelligent Transportation Systems (ITS) applications.

The methodology begins with preprocessing traffic data collected from a 25-kilometer highway section. Data sampled at five-minute intervals undergoes noise reduction and normalization to enhance computational efficiency. AOSVR employs a Radial Basis Function (RBF) kernel, with hyperparameters $\gamma=118$ and $\sigma=16$, optimized for balancing margin maximization and generalization.

To validate the model's performance, the study compares AOSVR against the Multi-Layer Artificial Neural Network (MLANN). The results reveal that AOSVR outperforms MLANN across all metrics, with RMSE reduced from 12.44% to 5.82%, MARE from 4.26% to 1.84%, and RME from 0.82% to 0.49%. This demonstrates the model's robustness in capturing nonlinear patterns and adapting to fluctuating traffic conditions. However, AOSVR incurs a higher computational cost during training (30-50 seconds) compared to MLANN's 6 seconds, attributed to the quadratic programming requirements of SVR. The paper concludes that AOSVR is a superior tool for short-term traffic prediction, offering higher accuracy and adaptability for ITS deployment.

3 Methodology

This methodology describes the preprocessing and modelling techniques employed to predict short-term traffic patterns in a 5G network using GRU, LSTM, and SVM models. The steps include data preprocessing, feature engineering and model implementation.

Previous studies have primarily focused on the use of LSTM to predict long-term traffic patterns, this research will aim to explore the use of both GRU and LSTM and compare their performance to SVR, in the prediction of 5G short-term traffic where limited research has been done.

3.1 Data Collection

The dataset that was used for this study was collected from telecommunication Operator A. It features hourly 5G traffic data that spans for a period of 1 week. The data contains a total of 176,544 samples with two primary attributes:

- **Time:**
 - This was captured in the format *dd/mm/yyyy h:mm* and indicates the exact hour that traffic was collected.
- **Traffic:**
 - This describes the volume of traffic in a continuous numerical format at the corresponding timestamp. The traffic values range from 0.00 to 590.54 GB, with a mean of 7.35 and a standard deviation of 13.28, highlighting significant variance in the data.

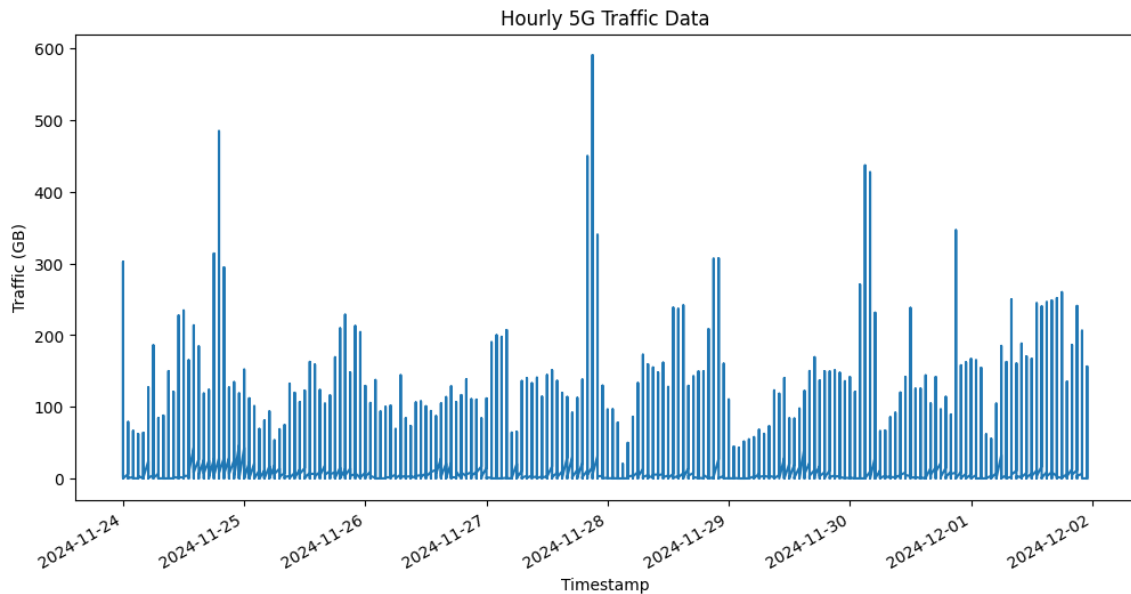


Figure 3.1 5G Traffic Data

The traffic data exhibited a wide range of values with occasional extreme peaks, regarded as outliers. Figure 3.1 shows the hourly 5G data used for this study. The nature of the data is continuous, with random values at different times. There is a pattern also observed in the data, traffic generally is seen to decrease at early hours in the morning then increases between 12:00 PM – 7:00 PM. These are known to be busy hours since they have the highest volume of traffic. The traffic then falls back again during the night. The data is also suitable for time-series modelling.

3.2 Feature Engineering

The traffic data used in this research is a time-series sequential data, values at time t depend on previous values at time $(t-1, t-2...)$ etc. Machine learning models, like SVM, require independent and identically distributed (i.i.d.) data. To use time-series data in the SVM model, lagged features were created to provide past values as inputs (features) and the future value as the target (output). A look_back variable is defined in the model implementation to choose the number of past steps to include as features. For example, look_back = 3 means using $x_{t-1}, x_{t-2}, x_{t-3}$ to predict x_t . A feature matrix X and target vector y were also defined where each row contains traffic values from the previous look_back time steps and each entry corresponds to the traffic value at the next time step, respectively.

In the case of the SVM model, the feature matrix X remains 2D in the form $(samples, look_back)$. For the GRU/LSTM models, the feature matrix is reshaped into a 3D tensor with dimensions $(samples, look_back, features)$. GRU/LSTM models naturally learn sequential dependencies, but they still require properly formatted data. Lagged features ensure the data is structured in a way that reflects temporal relationships. As for SVM, it does not natively handle sequences. In their case, lagged features encode the sequential dependency explicitly, allowing the model to learn patterns based on past values.

3.3 Model Implementation

Three different models were applied in this study Gated Recurrent Units (GRU), Long Short-Term Memory (LSTM) and Support Vector Machines (SVM).

3.3.1 Long Short-Term Memory (LSTM) Model

The Long-Short Term Memory model is a type of Recurrent Neural Network (RNN) (Greff *et al.*, 2016) made up of memory cells and gates. Figure 3.1 illustrates the architecture of the LSTM model.

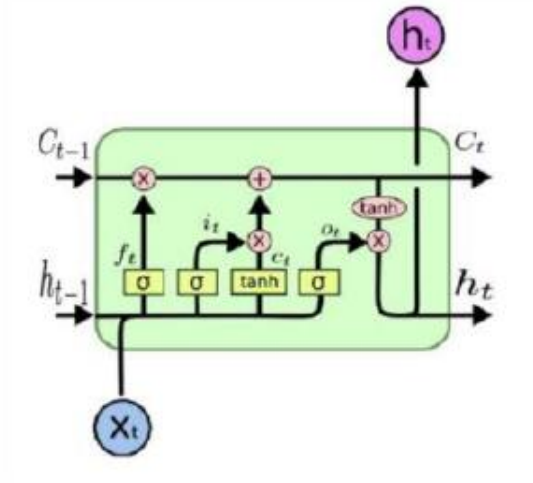


Figure 3.1 LSTM Architecture

The LSTM cell consists of three main elements:

- Memory Cell $C(t)$ that stores information over time and acts as a conveyor belt, allowing the network to preserve important information across long time intervals.

- Hidden State $h(t)$ that represents the output of the LSTM cell at a given time step and is used as input to the next time step or final output of the model.
- Three gates where each gate is a fully connected layer with a sigmoid activation function (σ) that outputs a value between 0 which means to block and 1 which means to allow:
- The Forget Gate decides which information to remove from the memory cell defined by below Equation 1:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

Where W_f describes the weight matrix for the forget gate, b_f is the bias vector for the forget gate, h_{t-1} is the hidden state from the previous time step, x_t is the input at the current time step and σ is the sigmoid activation function defined by Equation 2 below:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (2)$$

- The Input Gate i_t decides which new information to store in the memory cell defined by Equation 3:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

Where W_i is the weight matrix for the input gate and b_i is the bias vector for the input gate.

- The Output Gate decides which information to pass as the hidden state defined by Equation 4:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

Where W_o is the weight matrix for the output gate and b_o is the bias vector for the output gate.

- The Hidden State h_t is computed using the output gate and the memory cell defined by Equation 6:

$$h_t = o_t \odot \tanh(C_t) \quad (6)$$

Where $C_t = t \odot C_{t-1} + i_t \odot \tilde{C}_t$

The memory cell allows LSTM to retain information from earlier in the sequence, making it effective for long-term dependencies. In addition, the gating mechanism ensures gradients flow effectively during backpropagation, addressing a major limitation of traditional RNNs. This makes LSTM suitable for time-series analysis with varying trends and patterns. Applications where LSTMs are widely used are time-series forecasting, speech recognition, and natural language processing.

3.3.2 Gated Recurrent Units (GRUs)

Gated Recurrent Units like LSTMs are also a type of RNN (Dey and Salem, 2017), but unlike LSTMs they have a simpler structure with no memory cells and fewer gates. This makes GRUs computationally more efficient while still addressing the vanishing gradient problem. Below Figure 3.1 illustrates the architecture of GRUs.

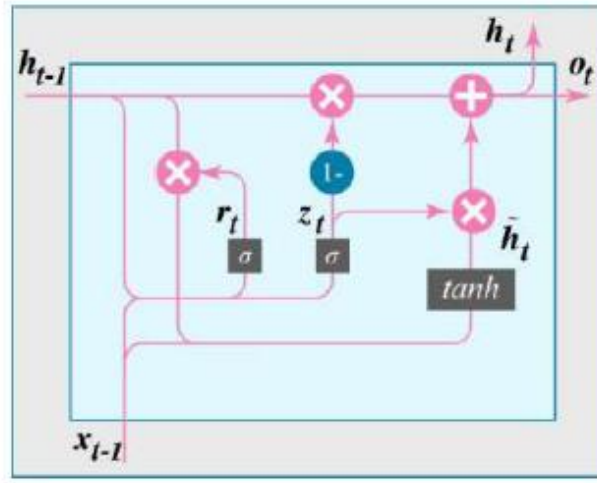


Figure 3.1 GRU Architecture

The GRU processes sequential data by controlling the flow of information using two key components:

- a. Update Gate (z_t):

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (7)$$

Where z_t is the output of the update gate, which is a vector with values between 0 and 1. It decides how much of the past information to retain and how much of the new information to update. W_z is the weight matrix for the update gate, b_z is the bias vector for the update gate, h_t is the concatenated vector of the previous hidden state h_{t-1} and the current input x_t and σ is the sigmoid activation function.

- b. Reset Gate (r_t):

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (8)$$

Where r_t is the reset gate output, and its values range between 0 and 1, it determines how much of the past information to forget for the current computation. Unlike LSTM, GRU does not have a separate

memory cell. Instead, it directly updates the hidden state. W_r is the weight matrix for the reset gate and b_r is the bias vector for the reset gate.

c. Candidate Hidden State ($\tilde{h}_t$):

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \quad (9)$$

Here $\tilde{h}_t$ is the candidate hidden state, representing the new information computed at the current time step, W_h is the weight matrix for generating the candidate hidden state, b_h is the bias vector and $r_t \odot h_{t-1}$ is the element-wise product of the reset gate and the previous hidden state which determines how much of the past hidden state contributes to the candidate hidden state.

d. Current Hidden State (h_t):

$$h_t = z_t \odot (h_{t-1} + (1 - z_t) \odot \tilde{h}_t) \quad (10)$$

Where h_t is the current hidden state, a combination of the previous hidden state h_{t-1} and the candidate hidden state $\tilde{h}_t$ weighted by the update gate.

GRU models have fewer gates and no separate memory cells which makes them less computationally expensive in comparison to LSTMs. GRUs perform well on tasks with moderate-length dependencies and due to their smaller structure, the risk of overfitting is reduced, particularly with small datasets.

3.3.3 Support Vector Machines (SVMs)

Support Vector Machines (SVMs) are powerful supervised learning algorithms used for classification and regression tasks (Chandra and Bedi, 2021). In the context of this study, Support Vector Regression (SVR) is employed to predict 5G traffic data. Figure 3.1 below shows the basic architecture of SVMs.

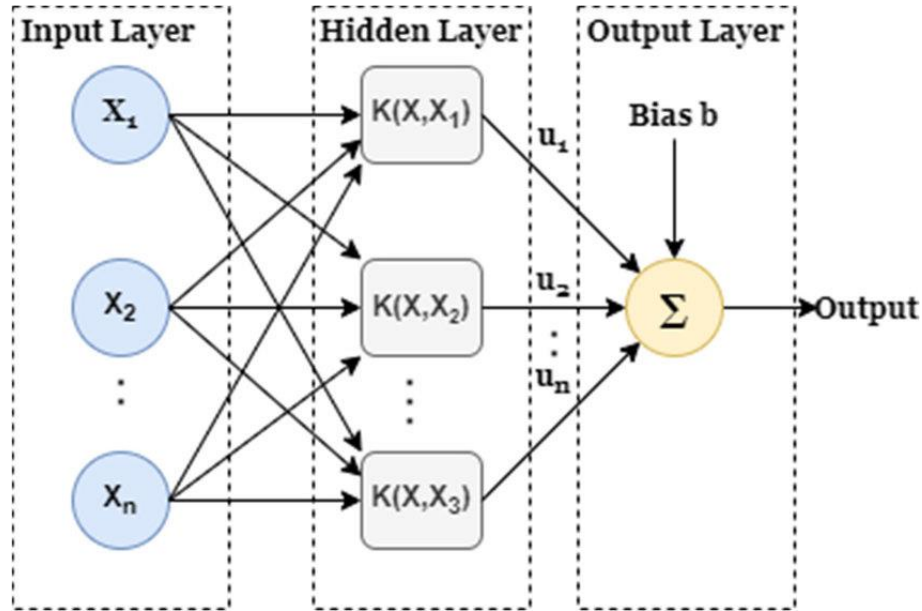


Figure 3.1. SVR Architecture

There are main components that make up SVR:

a. Kernel Function:

Maps input data into a higher-dimensional space to handle non-linear relationships. Common kernels include the Linear Kernel which works well for linearly separable data and the Radial Basis Function (RBF) which handles non-linear relationships by mapping data into infinite-dimensional space.

b. Hyperparameters:

C (Regularization Parameter) balances model complexity and prediction accuracy on the training set. Larger C focuses on minimizing error where there is higher complexity and to reduce the risk of overfitting. Smaller C simplifies the model by allowing some error in the case where that aim is to achieve better generalization. ϵ (Margin of Tolerance) specifies a margin of tolerance where predictions are not penalized for being close to the actual values. Larger ϵ are less sensitive to small deviations, leading to a simpler model while smaller ϵ focuses on minimizing all deviations. γ (RBF Kernel Coefficient) controls the influence of a single training sample. Smaller γ points farther away from the decision boundary to have more influence while larger γ points closer to the decision boundary.

SVR aims to find a function $f(x)$ that minimizes $\frac{1}{2} ||w||^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*)$ where $||w||^2$ is the regularization term to minimize model complexity, ξ_i, ξ_i^* are the slack variables for data points outside the ϵ -margin.

SVR is suitable to model both linear and non-linear relationships, making it flexible. The ϵ -margin also makes SVR more robust to the noise in data while the C parameter allows fine-tuning of the trade-off between error and generalization. The disadvantage however with SVR models is the computational time being high when dealing with large datasets.

Python programming language was used as the main language for the model's implementation. The train-test split method was implemented to ensure the models ability to generalise on unseen data. For this study, the dataset was split into a training set (80% of the total data) and a testing set (20%). The training set was used to train the models, while the testing set served as unseen data for evaluating their predictive performance. In machine learning, train-test splitting prevents data leakage, where information from the test set might inadvertently influence the model during training. This ensures that the model's performance reflects its ability to generalize to new data, rather than its capacity to memorize patterns from the training data.

Unlike standard machine learning tasks, time-series data has an inherent sequential structure. The value at any given time depends on previous values. As such, the order of the data must be preserved during splitting to mimic real-world scenarios where future data is predicted based on past observations.

To achieve this, *shuffle=False* was applied during the splitting process, ensuring that the training set consisted of earlier time steps and the testing set included only future time steps that followed the training set. Preserving the temporal order prevents data leakage. For example, using future values to predict past values (if shuffling were enabled) would create an unrealistic scenario and artificially inflate performance metrics. In addition, in a real-world 5G traffic prediction scenario, models are deployed to forecast future traffic patterns. By keeping the test set exclusively composed of future data, the evaluation aligns with how the model would perform in practical applications.

The TimeSeriesSplit method in Python's scikit-learn library was used for cross-validation with the SVM model. Unlike standard k-fold cross-validation, which randomly shuffles the data and splits it into folds, TimeSeriesSplit ensures that each training fold includes only past data relative to the validation fold and that the validation fold contains future data relative to the training fold.

A 5-fold split was applied, the first training fold includes the first 20% of the data, validation includes the next 20%. The second training fold includes the first 40%, validation includes the next 20%. This pattern continues until the entire dataset is covered. Time-series data is inherently sequential hence using future data to train the model (as in standard k-fold cross-validation) would violate the temporal dependency and result in data leakage. In practice, 5G traffic prediction models are deployed to forecast future traffic based on past observations. TimeSeriesSplit mirrors this workflow by ensuring that the validation set always follows the training set temporally. Furthermore, with each split, the training set size grows, allowing the model to learn from more data and improving its predictive performance.

For GRU and LSTM, a validation split was used instead of cross-validation, the training data (80% of the entire dataset) was further divided into a training subset (80%) and a validation subset (20%). The validation subset was used to tune hyperparameters such as the number of GRU/LSTM units, batch size, and learning rate. Unlike traditional machine learning models, GRU and LSTM inherently learn temporal dependencies. As a result, training on a fixed subset and validating on another subset is often sufficient for hyperparameter tuning.

The validation split was applied using TensorFlow's fit method. Validation split was set to 0.2 to reserve 20% of the training data for validation. Epochs were set to 10 to reduce the number of training iterations to balance accuracy and computational efficiency and Batch Size was set to 32 to control the number of samples processed before updating model weights.

For this study, RMSE and R-squared were chosen as the primary evaluation metric due to their robustness in penalizing larger errors and their ability to provide a clear interpretation of prediction quality. The RMSE is calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (11)$$

Where n are the number of data points in the test set, y_i is the actual observed traffic value at time i and $\hat{y}_i$ is the predicted traffic value at time i . RMSE measures the square root of the average squared differences between the predicted and actual values. These squared differences are aggregated, averaged, and then square-rooted to bring the error metric back to the original scale of the data.

R-squared is computed as follows:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (12)$$

Where y_i are the actual observed values, $\hat{y}_i$ are the predicted values from the model and $\bar{y}$ is the mean of observed values.

Squaring the errors before averaging gives higher weight to larger deviations between predicted and actual values. In 5G traffic prediction, extreme deviations (e.g., sudden spikes or dips) can have a significant impact on network performance. RMSE ensures that models are penalized heavily for failing to capture these anomalies, making it a critical metric for evaluating predictive accuracy. RMSE retains the same scale as the original target variable (traffic values in this case), making it easy to interpret. For instance, if $RMSE = 5$, it implies an average prediction error of 5 traffic units.

R-squared provides a simple and intuitive measure of the model's explanatory power, represented as a proportion. For instance, an R-squared of 0.70 indicates that the model explains 70% of the variance in the dependent variable, making it easy to understand the model's effectiveness.

4 Testing Results

4.1 Data Preprocessing

Due to the variance in data and outliers, sufficient preprocessing techniques were required to improve the model's performance and stability.

To overcome the noise in the data, a 10-point moving average was first applied to the raw traffic data. Each value was replaced with the average of its 10 neighbours (5 on each side). This aids in reducing short-term fluctuations that are caused by noise or measurement errors and it preserves long-term trends that are essential for time-series modelling. Since GRU and LSTM are sequential models, they rely on identifying patterns over time. Noise present in the data can obscure these patterns which may lead to either overfitting or instability during the training. The smoothing process ensures that the models focus on meaningful patterns rather than random fluctuations. Figure 4.1 shows the traffic data after 10-point moving average was applied.

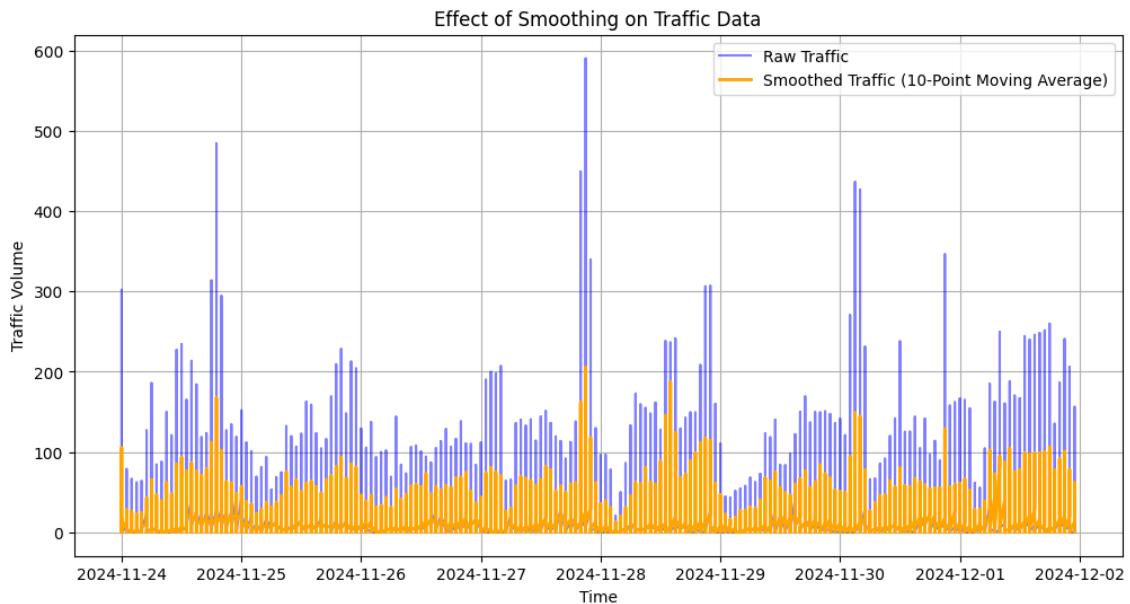


Figure 4.1 Smoothed Traffic Data

The 99th percentile method was then used to identify and handle the outliers in the traffic data. This method caps the extreme values that can disproportionately affect model performance, while preserving most of the dataset. The 99th percentile is the value below which 99% of the data lies. In this dataset, that would mean the top 1% of traffic values were considered outliers. The traffic values range from 0.00 to 590.54. In this instance, traffic values greater than 500 GB would distort the scale and the variance of the dataset. Hence, values greater than that threshold were capped and the outliers were replaced with a value more representative of the overall data distribution. Figure 4.2 shows data after capping was applied.

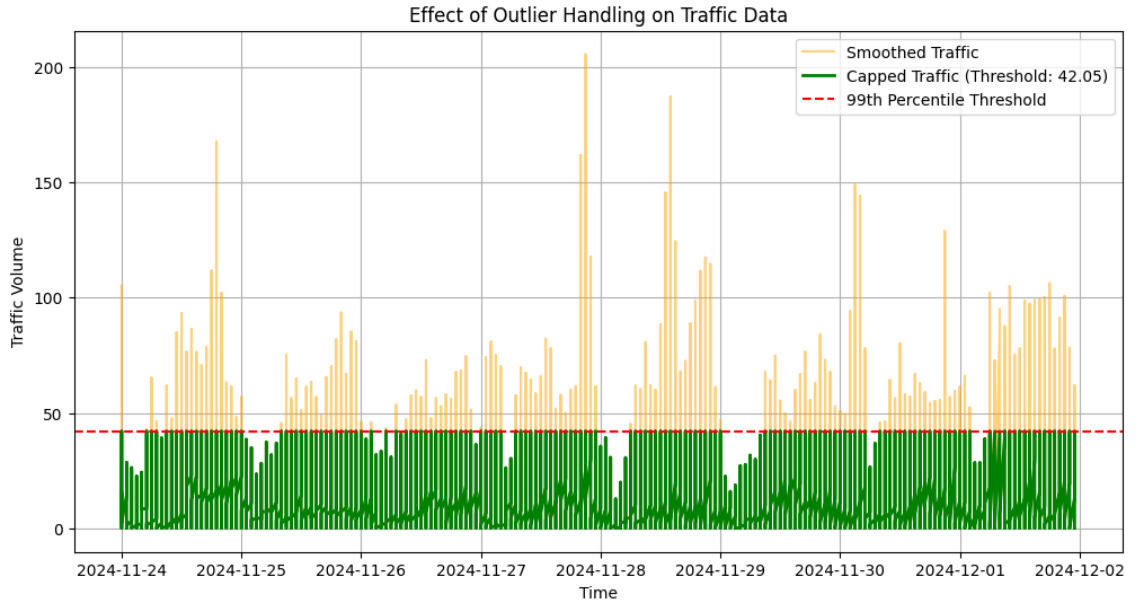


Figure 4.2 Capped Traffic Data

Instead of removing outliers completely, capping ensures that no data points are deleted, maintaining the temporal continuity required for time-series models. It also reduces the impact of extreme peaks on the models. Extreme values disproportionately affect gradient updates during training in GRU and LSTM models, leading to slower convergence and unstable predictions.

The next step was to normalize the traffic values range between 0 to 1, using Min-Max normalisation to standardise the input values for all models.

$$x = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (12)$$

This was done to ensure numerical stability during training, specifically required for neural networks. GRU and LSTM models are sensitive to the scale of input features. Large differences in feature magnitudes can lead to slower convergence or sub-optimal performance. Therefore, Min-Max normalization is ideal for time-series data where relative differences between values are more important than absolute magnitudes. Figure 4.3 shows traffic data after normalization was applied.

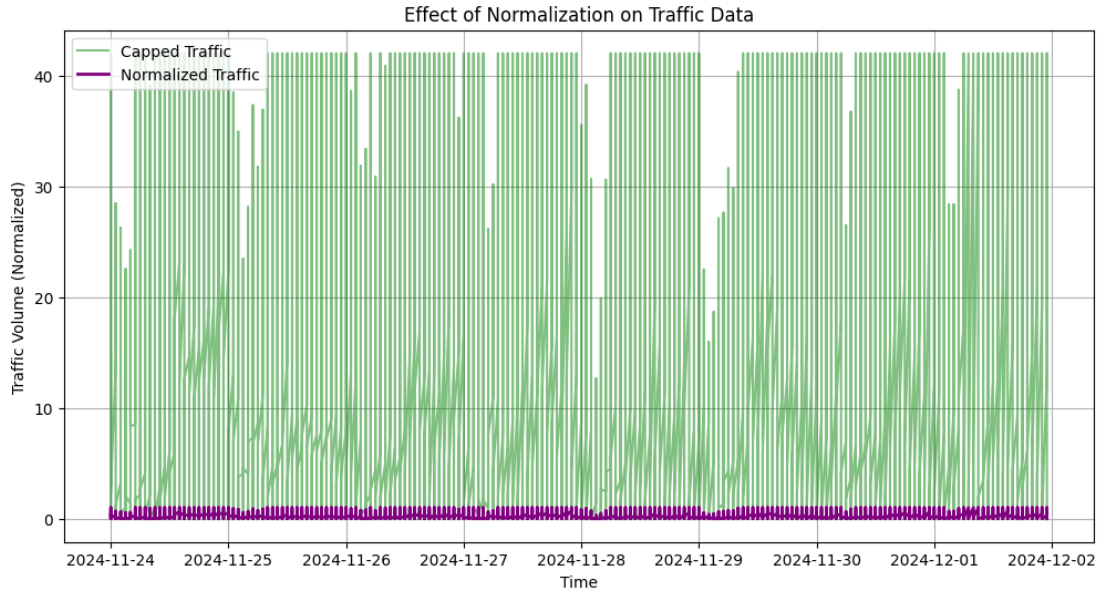


Figure 4.3 Normalised Traffic Data

4.2 Evaluation Results

The RMSE (Root Mean Squared Error) values for the models on the test set were computed as follows:

Model	RMSE
GRU	5.29
LSTM	5.32
SVM	5.36

Table 4.1. RMSE Results

The GRU model achieved the lowest RMSE (**5.29**), indicating that it performed slightly better in predicting traffic patterns on the test set. GRU's simpler architecture compared to LSTM may have contributed to its marginally better performance, as it requires fewer parameters and avoids overfitting for moderately complex sequences. GRU can efficiently learn short- to medium-term dependencies in the data, making it well-suited for this short-term traffic prediction task.

The LSTM model achieved an RMSE of **5.32**, just slightly higher than GRU. This suggests that LSTM's ability to capture longer-term dependencies may not have provided a significant advantage in this case. The performance difference between GRU and LSTM is negligible, which implies that both models are capable of accurately learning the temporal patterns in the traffic data. LSTM's more complex gating mechanisms (input, forget, and output gates) did not yield significantly better results. This might be because the dataset primarily exhibits short-term dependencies, which both GRU and LSTM can capture effectively.

The SVM model produced the highest RMSE at **5.36**, showing that it is less effective than GRU and LSTM for time-series prediction. SVM relies on lagged features to model temporal relationships explicitly, but it lacks the ability to learn sequential dependencies inherently like GRU and LSTM. While SVM can approximate the general trend of the traffic data, it struggles to model complex non-linear patterns and interdependencies in the sequence.

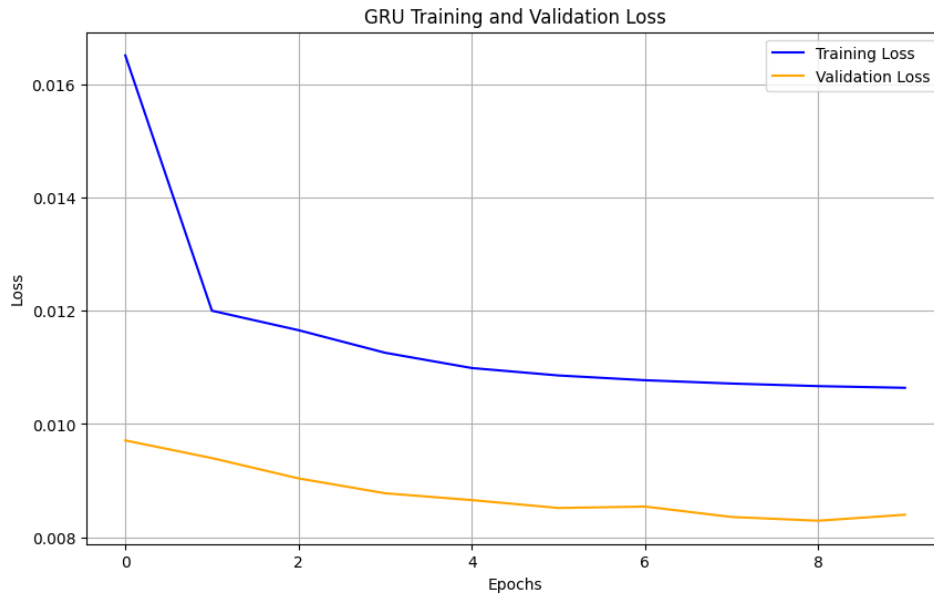


Figure 4.1

Figure 4.4. GRU Model Training and Validation Loss

The GRU model training loss (blue) decreases rapidly in the first epoch, starting at around **0.016** and stabilizing around **0.0105** by the 9th epoch while the validation loss (orange) also decreases but more gradually, stabilizing near **0.0083**. The gap between training and validation loss remains small throughout training. This indicates that the GRU model is learning effectively and generalizing well without signs of overfitting. Both the training and validation loss curves stabilize after epoch 6, suggesting that the model has reached its learning capacity. Further training beyond this point may not significantly reduce the loss.

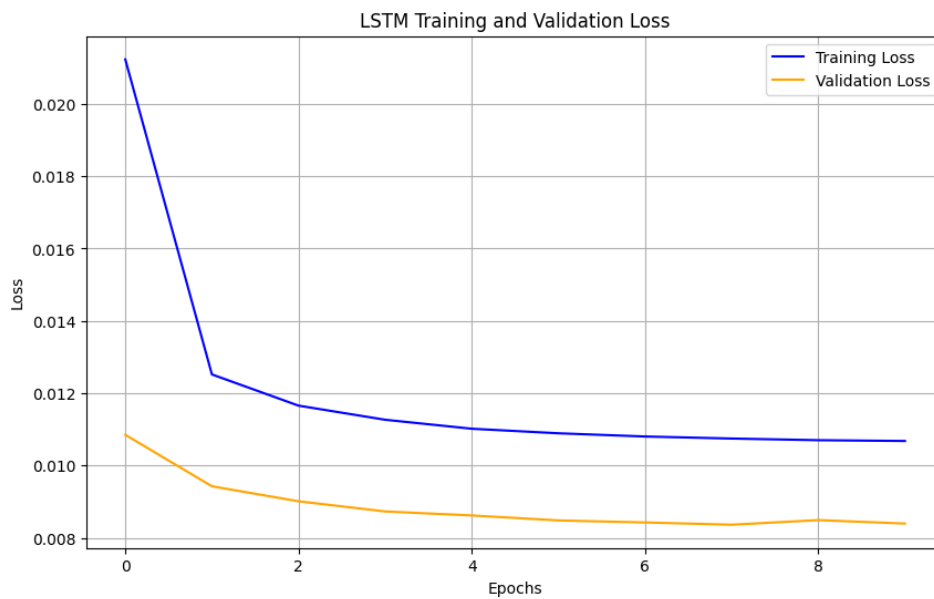


Figure 4.2. LSTM Model Training and Validation Loss

The LSTM model training loss starts higher at **0.021** and drops sharply to about **0.0115** within the first 2 epochs, stabilizing near **0.0103**. The validation loss also decreases quickly, reaching around **0.0083** by epoch 5 and stabilizing thereafter. The training and validation losses are close to each other throughout training, with the validation loss stabilizing earlier than the training loss. The validation loss also plateaus after epoch 5, similar to the GRU model. The minimal gap between training and validation loss suggests good generalization, like GRU. The LSTM model's validation loss also stabilizes slightly earlier than the GRU, indicating efficient learning of temporal patterns. The higher initial loss compared to GRU may reflect the additional complexity of the LSTM architecture, which requires more learning in the first few epochs.

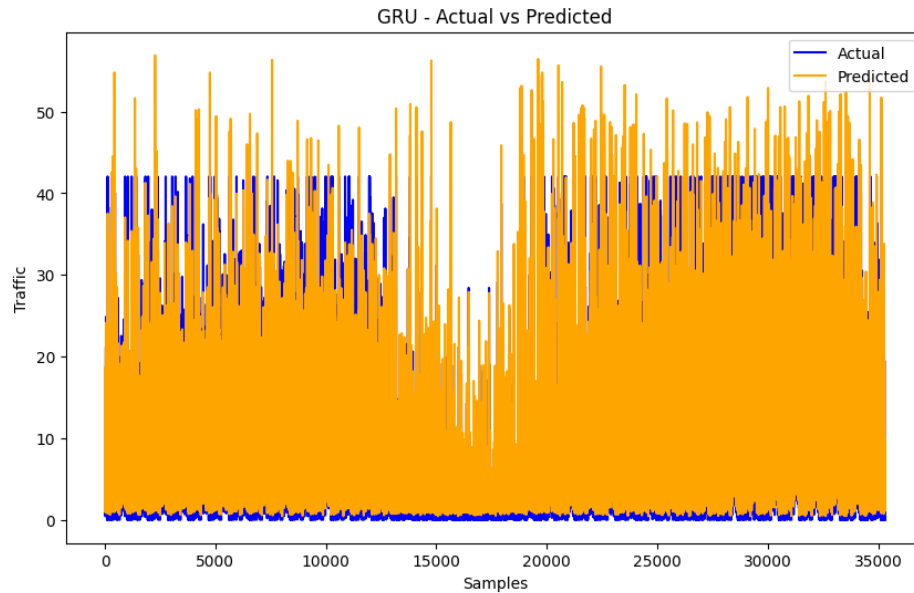


Figure 4.3. GRU Actual vs. Predicted Traffic Values

The GRU model predicts the general trend of the traffic data quite well. The predictions align closely with the actual values for most parts of the data, particularly during stable traffic periods. GRU struggles to capture the high variance and extreme peaks in the data. The model produces more smoothed predictions compared to the actual traffic, especially in regions where traffic spikes rapidly. For periods with lower traffic (e.g., near the beginning and end), GRU's predictions show reasonable alignment with the actual values. In the middle section of the graph, the predicted traffic shows more spread, deviating slightly from the actual values, especially during traffic dips. GRU captures the overall traffic trends efficiently and smooths out smaller fluctuations. However it fails to predict the rapid traffic spikes accurately, which indicates a challenge in modelling extreme short-term variations. This is due to the reason that GRU, being a simpler architecture compared to LSTM, might not fully capture longer-term dependencies or non-linear patterns in traffic spikes. Figure 4.3 shows actual vs predicted traffic values for the GRU model.

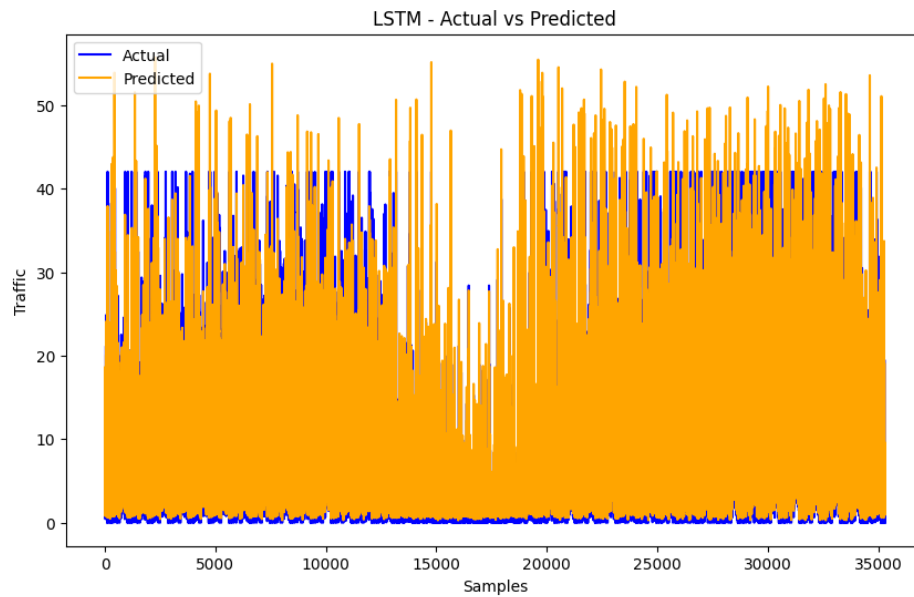


Figure 4.4. LSTM Actual vs. Predicted Traffic Values

Similar to GRU, the LSTM model captures the overall traffic trend quite well. The predictions align closely with the actual traffic, especially during stable traffic periods. The LSTM predictions display similar smoothing behaviour but seem slightly better at capturing some peaks compared to GRU. While the alignment with extreme values (spikes) is not perfect, the LSTM shows slightly less deviation in these regions. The LSTM model aligns well with low-traffic regions, in similarity to GRU, showing consistent predictions. In the middle section, The LSTM model behaves like the GRU, with smoothed predictions and occasional deviations during traffic dips. LSTM models the overall trend and slightly improves predictions for traffic spikes compared to GRU. However, like GRU, LSTM still smooths out extreme traffic variations and struggles to fully align with high peaks. That is because while LSTM is designed to capture long-term dependencies better, the traffic data likely has more short-term variations that are harder to predict.

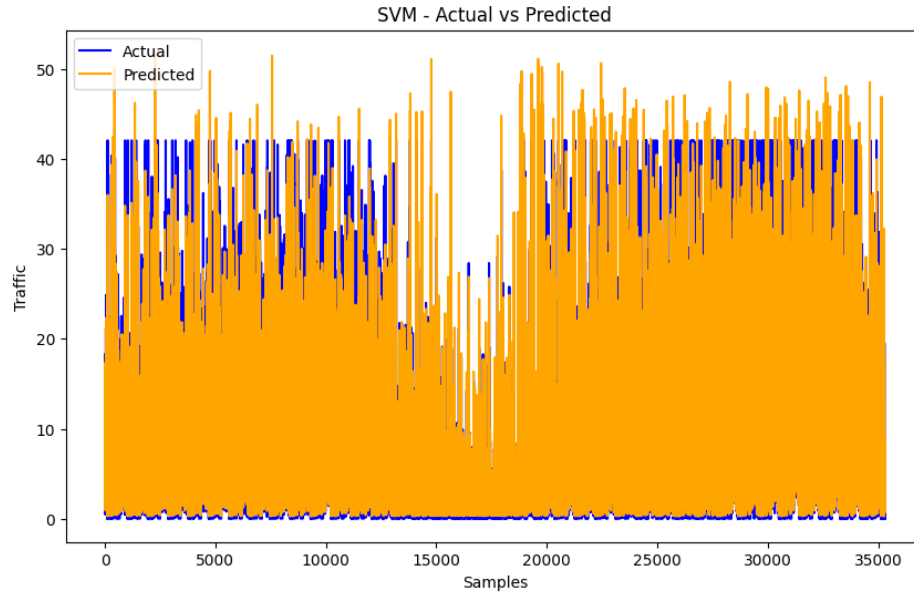


Figure 4.5. SVM Actual vs. Predicted Traffic Values

SVM captures the general traffic trend but performs less accurately compared to GRU and LSTM. The predicted values deviate more from the actual traffic during stable traffic periods. SVM predictions show significant smoothing and fail to capture the rapid peaks and dips in the traffic data. The high variance in the traffic data is not well modelled, and extreme spikes are significantly underrepresented. SVM predictions align moderately well with low-traffic regions but still exhibit noticeable smoothing. In the middle section, SVM predictions deviate more noticeably from actual values compared to GRU and LSTM, particularly during traffic dips and spikes. In general, SVM captures the general traffic trend and performs adequately for moderately varying regions. Nevertheless, the model struggles to capture traffic spikes and dips, producing overly smoothed predictions. That is due to the reason that SVM relies on lagged features for time-series prediction and lacks the sequential learning capabilities of GRU and LSTM, limiting its ability to capture complex temporal dependencies and non-linear relationships.

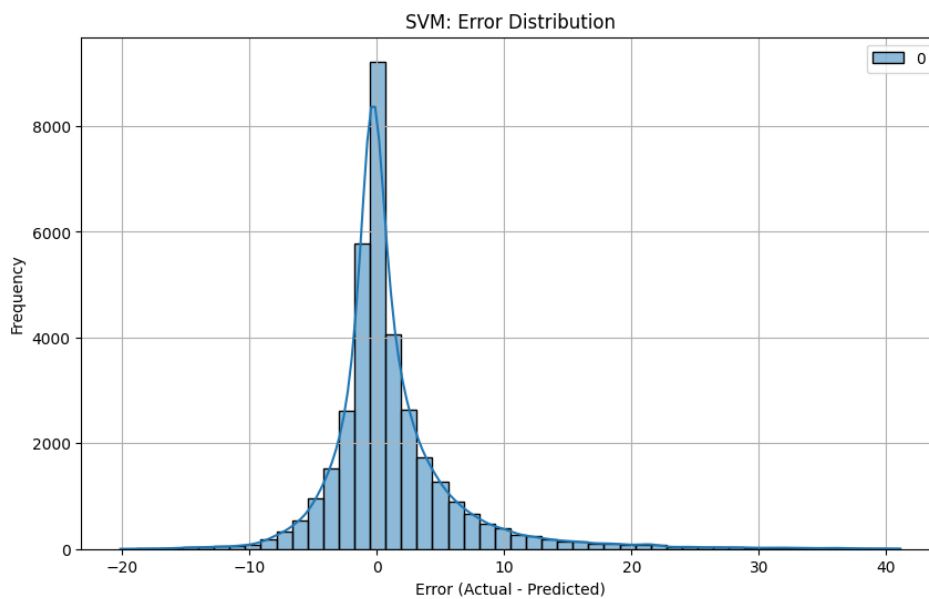


Figure 4.6. SVM Error Distribution

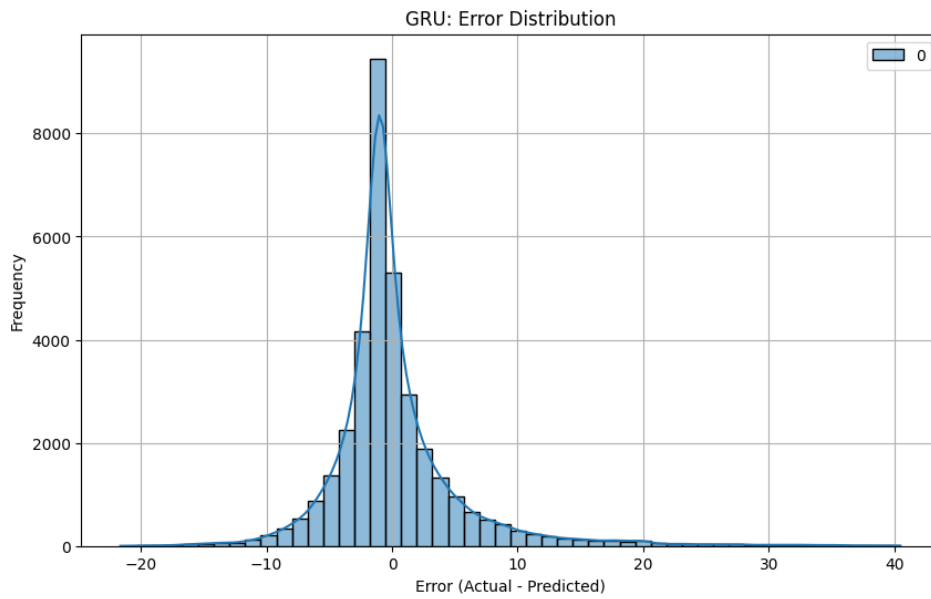


Figure 4.7. GRU Error Distribution

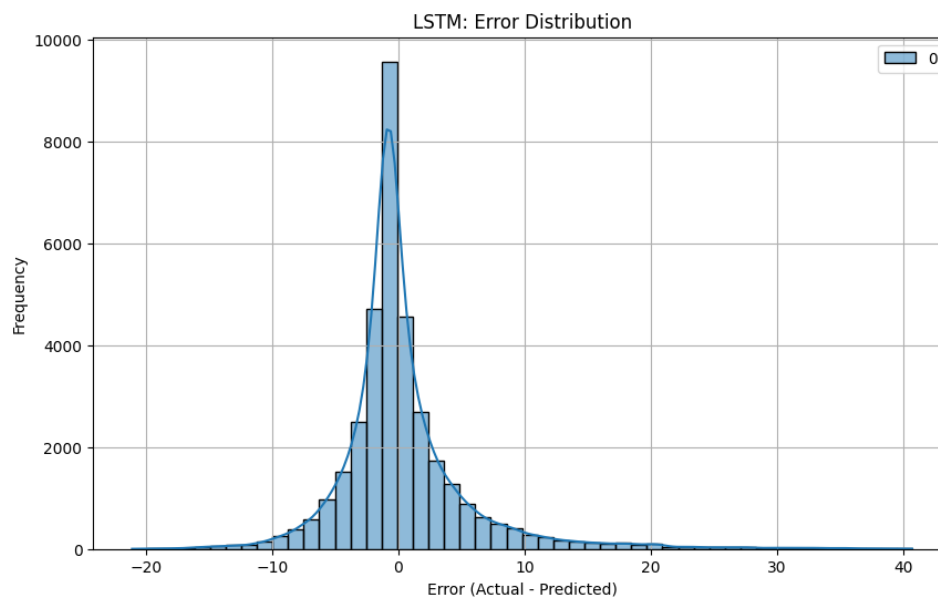


Figure 4.8. LSTM Error Distribution

The error distribution graphs for LSTM, GRU, and SVM models provide deeper insights into their predictive performance. LSTM exhibits a narrow, symmetrical error distribution centered around zero, indicating that most predictions are close to the actual values. It has fewer extreme errors, confirming its ability to handle high variance and traffic spikes better. In addition, the tighter spread reflects the LSTM's capability to learn complex temporal dependencies accurately. GRU also shows a similar error distribution to LSTM, with a sharp central peak and symmetrical spread around zero. However, its errors are slightly more spread out, with a marginally higher frequency of large deviations. This reflects GRU's computational efficiency while maintaining strong predictive performance. The SVM error distribution is wider and right skewed, with more frequent positive errors (under-predictions). Furthermore, SVM struggles to capture sharp spikes and dips in traffic, resulting in larger deviations and smoothing out extreme values, its wider error spread highlights its limitations compared to LSTM and GRU.

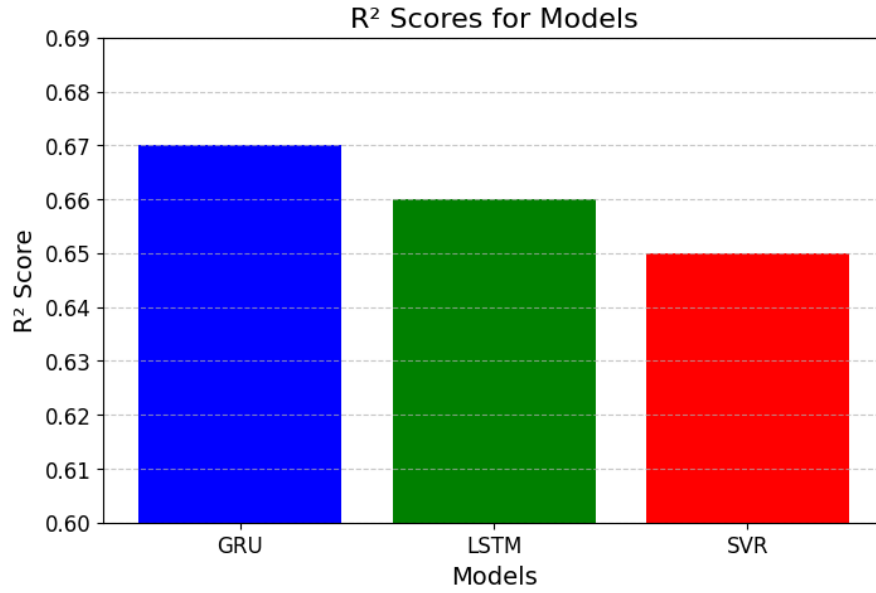


Figure 4.9. R^2 Scores of GRU, LSTM and SVR

R^2 scores for GRU, LSTM and SVR were **0.67**, **0.66** and **0.65** respectively as illustrated in Figure 4.9. These results indicate a moderate level of predictive accuracy for short-term traffic forecasting in 5G networks. The similarity in R^2 scores across GRU, LSTM, and SVR suggests that all three models are comparably effective for the dataset used. This indicates that each model, despite its unique architecture and approach, has captured the underlying traffic patterns to a similar extent. However, the small differences may hint at slight advantages in GRU's computational efficiency.

In comparison to (Gao, 2022), the R^2 scores are less where an average value of 0.89 for LSTM and GRU was achieved. The study introduced a Smoothed Long Short-Term Memory (SLSTM) model for predicting 5G network traffic. Major differences are found in the type of dataset used, the previous study uses 5G bandwidth traffic data (static and dynamic conditions), which may exhibit distinct patterns compared to the dataset used in this study, particularly in terms of periodicity or stationarity. Furthermore, the SLSTM incorporates a stabilization process and adaptive mechanisms for layer updates based on prediction accuracy, which may enhance its performance. SLSTM also dynamically adjusts layers and units, likely optimizing the model's performance in real-time.

5 Conclusion

The comparative analysis of GRU, LSTM, and SVM models for short-term traffic prediction in 5G networks has revealed key insights regarding their performance and limitations. Each model was evaluated based on its ability to capture overall traffic trends, handle high variance, and predict traffic spikes and dips. The GRU model effectively captured the general traffic trends while offering a computationally efficient solution. It performed well on low-traffic regions but struggled with sudden traffic spikes, smoothing out these extreme variations. The LSTM model exhibited similar performance to GRU, with a slight edge in handling high variance and traffic peaks. Its ability to model longer-term dependencies was advantageous in scenarios where subtle patterns existed across time. The SVM model, while simple and interpretable, underperformed compared to GRU and LSTM. It smoothed out both traffic spikes and dips excessively, highlighting its inability to inherently model sequential dependencies.

Both GRU and LSTM models demonstrated strong performance in general trend prediction. Their sequential architectures allowed them to learn temporal dependencies in traffic data, making them superior to SVM. The SVM model's reliance on explicit feature engineering (lagged features) limits its ability to learn complex temporal patterns and non-linear relationships, leading to oversimplified predictions. For practical applications, GRU and LSTM offer robust solutions for traffic prediction in 5G networks. GRU, with its simpler architecture, can be preferred for real-time prediction tasks where computational efficiency is critical. The slightly better performance of GRU (0.67) could indicate its suitability for short-term, real-time predictions due to its efficient architecture. While SVM can serve as a baseline, it lacks the sophistication required for high-variance time-series forecasting.

Future work can explore hybrid models combining GRU/LSTM with SVM or ensemble methods to capture both trends and extreme variations.

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Appendix A

1G - 1st Generation

2G - 2nd Generation

3G - 3rd Generation

4G - 4th Generation

5G - 5th Generation

AI - Artificial Intelligence

AOSVR - Accurate Online Support Vector Regression

ARIMA - Auto-Regressive Integrated Moving Average

BP - Back Propagation

DGP - Deep Gaussian Process

EM - Expectation-Maximization

FFNN - Feed-Forward Neural Networks

GRU - Gated Recurrent Unit

ITS - Intelligent Transportation Systems

LSTM - Long Short-Term Memory

MAPE - Mean Absolute Percentage Error

MLANN - Multi-layer Artificial Neural Network

NRMSE - Normalized Root Mean Square Error

QoS - Quality of Service

RBF - Radial Basis Function

RMSE - Root Mean Square Error

SLSTM - Smoothed LSTM

SVM - Support Vector Machine

SVR - Support Vector Regression

WP - Wavelet Packet

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